

Physical Risk Assessment

2026-10-02 · 1 site(s)

Chattogram cement plant

Cement plant · Chittagong BGD · 22.3569, 91.8100

HIGH Chattogram cement plant (cement plant): overall physical exposure HIGH. Main exposures: tropical cyclone wind (HIGH); heat (land-cover indicator) (MODERATE). Water supply context: critical water stress.



1 · How the site sits

Chattogram cement plant sits at 35 m above sea level on flat ground (slope 5.7°), in built-up land cover (88% built-up within 250 m); the nearest surface water is about 206 m away; 0.4 km² of land drains towards it from up to 28 m higher.

Elevation	35 m a.s.l.	AWS Terrain Tiles ~10 m
Terrain position	flat	AWS Terrain Tiles
Slope at the site	5.7 °	AWS Terrain Tiles
Catchment draining to the site	0.41 km²	pysheds D8 flow routing
Height of upslope land above the site	28 m	AWS Terrain Tiles
Land cover at the site	built up	ESA WorldCover 2021
Built-up land within 250 m	88 %	ESA WorldCover 2021
Tree cover within 1 km	47 %	ESA WorldCover 2021
Nearest surface water	206 m	Sentinel-2 (2025-11-28)
Land within 500 m that has been under water in the radar record	0 %	Sentinel-1 (2024-10-29 → 2026-09-30)
Location	22.3569, 91.8100 — Chittagong, BGD	

2 · Hazard zones

Hazard	Rating	Where the site sits
River flooding	NEGLIGIBLE	Outside the modelled river floodplain up to a 1-in-500-year flood. Frequencies are historical; no climate-change uplift is applied to river floods.
Coastal flooding and sea-level rise	NEGLIGIBLE	Site is above 30 m; coastal flooding is not a hazard here.
Surface-water and observed flooding	NEGLIGIBLE	No significant surface-water flooding exposure found (Terrain flood routing rating: NEGLIGIBLE; Terrain position: flat; Upstream drainage area: 0.4052 km²).
Tropical cyclone wind	HIGH	Inside a tropical-cyclone region: the 1-in-100-year wind is about 55 m/s (Cat 3). Frequencies are historical (1980–2025); no climate-change uplift is applied.
Landslide and debris flow	LOW	Terrain around the site gives low susceptibility to slope failure or debris flow.
Wildfire	NEGLIGIBLE	No significant wildfire exposure found (Tree / shrub / grass cover within 1 km: 47.9 %; Tree cover within 1 km (crown-fire fuel): 47.3 %; Mean NDVI of surrounding area: 0.477).
Heat (land-cover indicator)	MODERATE	Moderate heat exposure (Built-up cover within 1 km: 51.0 %; Tree canopy within 1 km: 47.3 %; Latitude band: 22.4° (warm climate)). Indicator from land cover and latitude, not a temperature projection.
Water stress (context)	CRITICAL	Catchment water risk is Extremely High (4-5) (WRI Aqueduct 4.0, Chittagong). Water stress under business-as-usual: 2019 3.08, 2030 3.14, 2050 3.32, 2080 3.18

3 · Construction year and age

Built 1994; by 2050 it will be 56 years old, beyond the upper end of the typical design life (50 years). Major refurbishment or replacement is likely before then, which is the cheapest moment to build in the protection below.

4 · What will go wrong

Parts of this kind of site that matter most: kiln and preheater tower, raw and cement mills, clinker and cement silos, conveyors, electrical substations and motor control rooms, quarry and haul roads, dispatch by road, rail or jetty.

Tropical cyclone wind — **HIGH**

Inside a tropical-cyclone region: the 1-in-100-year wind is about 55 m/s (Cat 3).

The preheater tower, stacks and silos carry the highest wind loads. Roof sheeting, cladding and conveyor galleries fail first and loose items become missiles. Loss of grid power during a storm trips the kiln.

Because of its age: Roof fixings, cladding, sheeting and door seals loosen and corrode with age, and older structures were designed to the wind records and code of their time. (typical ageing mechanism; confirm with a condition survey)

Heat (land-cover indicator) — **MODERATE**

Moderate heat exposure (Built-up cover within 1 km: 51.0 %; Tree canopy within 1 km: 47.3 %; Latitude band: 22.4° (warm climate)). Indicator from land cover and latitude, not a temperature projection.

High ambient temperature derates mills, fans and electrical equipment, reduces clinker-cooler capacity and raises heat stress for workers around the kiln.

Because of its age: Cooling and ventilation plant is sized to historical design temperatures and loses capacity with age. (typical ageing mechanism; confirm with a condition survey)

Landslide and debris flow — **LOW**

Terrain around the site gives low susceptibility to slope failure or debris flow.

Quarry benches, haul roads and spoil heaps are the slope-sensitive parts: heavy rain destabilises faces and washes out roads, and debris reaching the plant platform blocks drainage and access.

Because of its age: Retaining structures, slope drainage and cut-slope protection deteriorate with age. (typical ageing mechanism; confirm with a condition survey)

Water stress (context) — **CRITICAL**

Catchment water risk is Extremely High (4-5) (WRI Aqueduct 4.0, Chittagong).

The plant needs water for cooling, dust suppression and concrete production; high stress raises the chance of abstraction limits in dry years.

Because of its age: Older plant tends to be less water-efficient and harder to retrofit. (typical ageing mechanism; confirm with a condition survey)

5 · What to do

Tropical cyclone wind — HIGH · Immediately

- Design or retrofit the building envelope to the local cyclone wind code for the modelled 1-in-100-year wind (roof tie-downs, rated cladding and fixings, impact-rated openings).
- Secure or remove rooftop plant and loose external items; inspect roof fixings before each season.
- Prepare a cyclone season plan: 72-hour shutdown procedure, backup power and fuel, staff shelter.
- Review windstorm insurance limits against the worst historical storm at this site.
- Commission a condition survey of the relevant structures and systems (age-related deterioration).

Heat (land-cover indicator) — MODERATE · Within 12 months

- Apply high-albedo cool-roof coatings to large roof areas (cuts roof surface temperatures significantly).
- Increase shade and tree canopy on site; low canopy cover amplifies local heat.
- Re-rate HVAC and cooling for higher design temperatures; add heat-stress protocols for outdoor staff.
- Commission a condition survey of the relevant structures and systems (age-related deterioration).

Water stress (context) — CRITICAL · Immediately

- Map the site's water sources and consumption; secure a second source or on-site storage for dry years.
- Commission a condition survey of the relevant structures and systems (age-related deterioration).

Basis: Ratings describe hazard intensity at the site against stated thresholds; they are not damage estimates. Flood depth rating (mean over a 250 m footprint at a 1-in-100-year event): below 0.05 m negligible, below 0.3 m low, below 1 m moderate, below 2 m high, above that critical; one level lower if the local defence standard is 1-in-100 or better. Screening-grade: it prioritises sites for survey and does not replace a hydraulic flood model, a structural condition survey or a geotechnical investigation. Terrain comes from a surface model: buildings and trees can raise apparent ground height. Sources: AWS Terrain Tiles (Terrarium, ~10 m) — s3://elevation-tiles-prod; Sentinel-2 L2A COGs — AWS Open Data via Earth Search STAC; ESA WorldCover 2021 v200 — s3://esa-worldcover; NOAA IBTrACS v04r01 tropical-cyclone tracks 1980–2025 (validated vs NOAA ISD station winds); Sentinel-1 RTC radar — Microsoft Planetary Computer; JRC CEMS-GloFAS river flood hazard maps v2.1; Deltares Global Flood Maps (Planetary Computer); FLOPROS v1 (Scussolini et al. 2016); WRI Aqueduct 4.0